Name : Alezay Salman

Roll no. : 25L-0558.

Assembly Language Worksheet

Flags: Carry (CF) · Zero (ZF) · Sign (SF) · Overflow (OF)

A short worksheet to help you get comfortable reading flags by hand. Below are 14 tiny 8086 programs, each just 5–6 lines using 8-bit registers. For each one, work through the instructions in order and write down what CF, ZF, SF, and OF equal once the program finishes. A couple of things worth keeping in mind as you go: INC and DEC never touch the Carry Flag, and AND / OR always clear both CF and OF, no matter what the operands are. Some of these are more subtle than they look, so read each line carefully rather than pattern-matching to an earlier question.

Question 1

```
mov al, 04h
mov bl, 02h
mov cl, 01h
add al, bl
add al, cl
```

CF	ZF	SF	OF
0	0	0	0

Question 2

```
mov al, 0FFh
mov bl, 01h
mov cl, 00h
mov dl, cl
add al, bl
```

CF	ZF	SF	OF
1	1	0	0

Question 3

```
mov al, 7Fh
mov bl, 01h
mov cl, 00h
mov dl, cl
add al, bl
```

CF	ZF	SF	OF
0	0	1	1

Question 4

CamScanner

```
mov al, 0FFh
mov bl, 0FFh
mov cl, 00h
mov dl, cl
add al, bl
```

CF	ZF	SF	OF
1	0	1	0

Question 5

```
mov al, 03h
mov bl, 05h
mov cl, 00h
mov dl, cl
sub al, bl
```

CF	ZF	SF	OF
1	0	1	0

Question 6

```
mov al, 05h
mov bl, 05h
mov cl, 00h
mov dl, cl
sub al, bl
```

CF	ZF	SF	OF
0	1	0	0

Question 7

```
mov al, 80h
mov bl, 01h
mov cl, 00h
mov dl, cl
sub al, bl
```

CF	ZF	SF	OF
0	0	0	1

Question 8

```
mov al, 05h
mov bl, 0FBh
add al, bl
```

CamScanner

```
mov cl, 7Fh
inc cl
```

CF	ZF	SF	OF
1	0	1	1

Question 9

```
mov al, 02h
mov bl, 03h
sub al, bl
mov cl, 01h
dec cl
```

CF	ZF	SF	OF
1	1	0	0

Question 10

```
mov al, 05h
mov bl, 08h
mov cl, 00h
mov dl, cl
cmp al, bl
```

CF	ZF	SF	OF
1	0	1	0

Question 11

```
mov al, 80h
mov bl, 80h
mov cl, 00h
mov dl, cl
add al, bl
```

CF	ZF	SF	OF
1	1	0	1

Question 12

```
mov al, 0FFh
mov bl, 01h
mov cl, 00h
mov dl, cl
cmp al, bl
```

CamScanner

CF	ZF	SF	OF
0	0	1	0

Question 13

```
mov al, 80h
mov bl, 00h
mov cl, 00h
mov dl, cl
or al, bl
```

CF	ZF	SF	OF
0	0	1	0

Question 14

```
mov al, 05h
mov bl, 03h
sub al, bl
mov cl, 0FFh
inc cl
```

CF	ZF	SF	OF
0	1	0	0

CamScanner

Q1. Program:

```
[org 0x100]
    mov bqx, arr
    mov, ax ,0
    mov cx, 10
kk: mov dx, [bx]
    and dx, dx
    jnp skip
    inc ax
skip: add bx, 2
    dec cx
    jnz kk
mov ax, 0x4c00
  int 0x21.
arr : dw 1,2,3,4,5,6,7,8,9,10.
```

Q2. Program:

```
[org 0x100]
jmp start;
 array : db -3, 5,-10,4,6,-7,1
 size : db 7
 max : db 0
 start:
  mov cl, [size]
  movch, 0
  mov bx, array
l2: mov al, [bx]
    cmp al,0
    js abs-mag
back: cmp al, max
    jle l1
    mov [max],al
l1: inc bx
    dec cx
    jnz l2
mov ax, 0x4c00
  int 0x21
abs_mag: xor al, 0x FF
        inc al
        jmp back
```

Q3 Physical Addresses

(a)
```
  1 0 0 0 0
+ 0 0 4 3 6
-----------
  1 E 2 0 6 H → address
```

(b)
```
  1 2 3 4 0
+ 0 7 9 2 0
-----------
  1 9 C 6 0 H → address
```

(c)
```
    74'F00
  + 02123
  ---------
    770 23 H → address
```

(d)
```
  00000
+ 06727
---------
  06727 H → address
```

(e)
```
  1F'FFF0
+ 04336
---------
wraparound 1,04326,H → address
```

(f)
```
  10 800
+ 00100
---------
  10900 H → address
```

(g)
```
    ① ①1
    A B010
+ 0 FFFF
---------
  B B00FH → address.
```

c. 1002
d. 0001
e. E000

Q5. [5 Marks]

Consider the program written below. For this program being executed, the current contents of the CS register are 0FE2h and the physical address pointed to by CS:IP is 0FF29h. The size of the MOV instruction is 3 bytes. For this program, determine the current contents of a) the IP register, and b) the AX register.

```
[ORG 0x100]
MOV AX, 10
MOV AX, 6
MOV AX, 5
MOV AX, 2
MOV AX, 1
MOV AX, 0x4C00
INT 0x21
```

Q4. Range of addresses.

(a)
$$\begin{array}{r} 10000 \rightarrow \text{start/1st address} \\ +\ 0FFFF \\ \hline 1FFFFH \rightarrow \text{last address.} \end{array}$$

(b)
$$\begin{array}{r} {}^{1}{}^{1}{}^{1} \\ 0FFF0 \rightarrow 1^{st}\ \text{address} \\ +\ 0FFFF \\ \hline 1FFEF \rightarrow \text{last address.} \end{array}$$

(c)
$$\begin{array}{r} {}^{1}{}_{1}{}^{1} \\ 10020 \rightarrow 1^{st}\ \text{address} \\ 0FFFF \\ \hline 2001FH \rightarrow \text{last address} \end{array}$$

(d)
$$\begin{array}{r} {}^{1}{}_{1}{}^{1} \\ 00010 \rightarrow 1^{st}\ \text{address} \\ +\ 0FFFF \\ \hline 1000FH \rightarrow \text{last address} \end{array}$$

(e)
$$\begin{array}{r} E0000 \rightarrow 1^{st}\ \text{address.} \\ +\ 0FFFF \\ \hline EFFFFH \rightarrow \text{last address} \end{array}$$

CamScanner

Q5. (a) contents of IP register.

$$\begin{array}{r} 0FF29 \rightarrow \text{physical address} \\ -\ 0FE20 \rightarrow \text{CS segment} \\ \hline 0009H \rightarrow \text{IP offset} \end{array}$$

(b) Contents of AX register

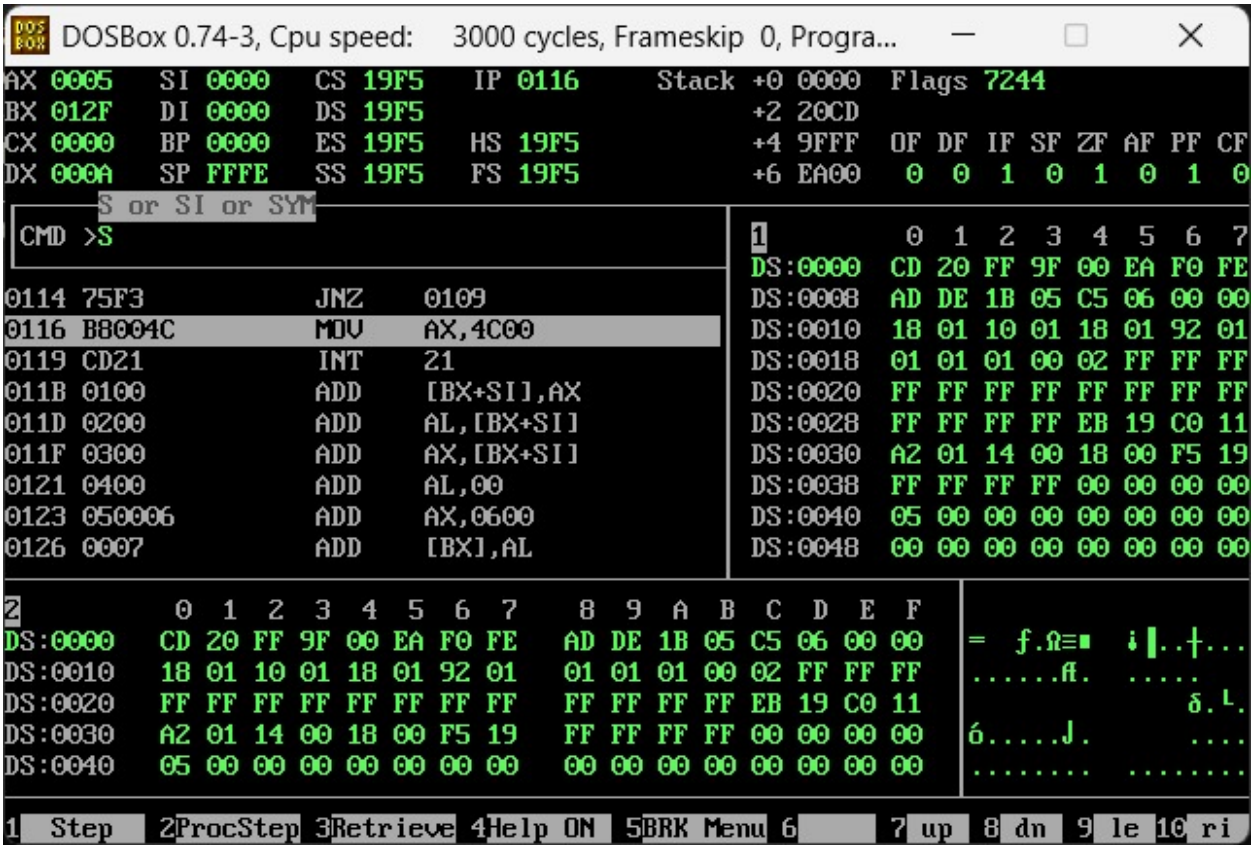
As IP points to next instruction to be executed, we need value of AX before IP.

$\Rightarrow$

MOV AX,10 → 102
MOV AX,6 → 105
MOV AX,5 → 108 → current value of A
MOV AX,2 → 10B

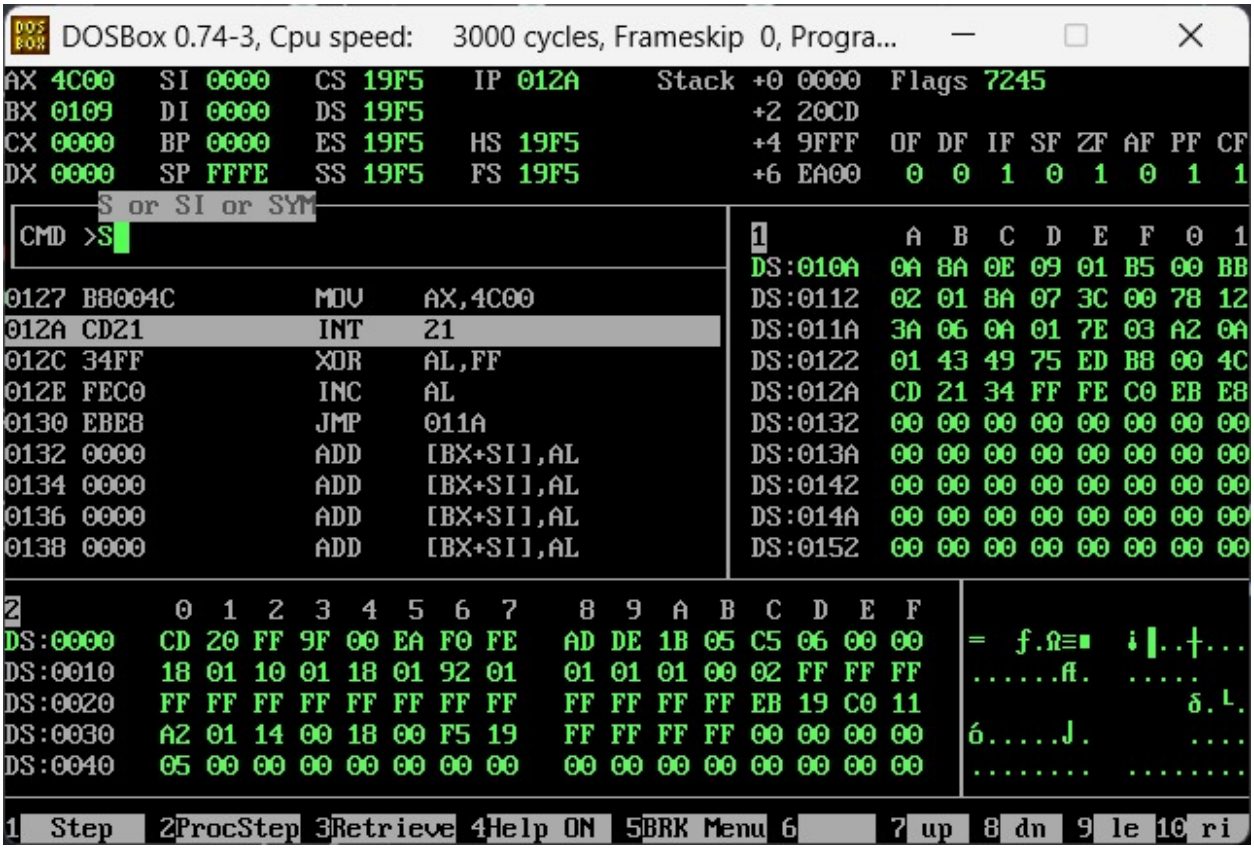
$\Rightarrow$ AX is at 0005H.

Q1



DOSBox execution screenshot (verification of program output)

Q2



DOSBox execution screenshot (verification of program output)